MILITARY SPECIFICATION

TEST HARNESS, DIRECTION FINDER SET

AN/ALM-28(*)

This Specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall be used for the subsystem performance testing and fault isolation of the Direction Finder Set AN/ALD-2.

1.2 Classification - The Test Harness, Direction Finder Set shall be Class 3 equipment in accordance with Specification MIL-T-21200 and shall consist of the items listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

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(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size or circuitry to be included therein shall be approved by the contractor. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction -

3.2.1 General - Unless otherwise specified herein, the equipment shall conform with all the applicable requirements of Specification MIL-H-15362 for design, construction and workmanship.

3.2.2 Enclosure - The equipment shall be provided with two combination cases and detachable covers in accordance with Specification MIL-T-21200.

3.2.2.1 Total Weight - The total weight of the equipment, including its covers and contents, shall be a minimum consistent with good design and shall not exceed 64 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible to operator during normal use.

3.2.3.2 Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be determined by the manufacturer and submitted to the procuring activity.

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3.2.4 Cables and Connector - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with operating controls. All internal wiring shall be AWG-22 (Minimum) in accordance with MIL-W-76 except as noted herein. For mechanical strength external cables shall be fabricated from wire AWG No. 20 minimum in accordance with MIL-W-76. Shielded cable shall be in accordance with MIL-C-7078. Identification shall be in accordance with MIL-W-5088.

3.2.4.1 Cable Assemblies, Power Electrical - Applicable input power cable and connection shall conform to MS-21993. Power leads from connector to cabinet shall be 6 feet (minimum).

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181. (See 3.3.5)

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient 30° ± 10°C
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V ac, 400 ± 20 cps 28 ± 0.5 V dc

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

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3.2.11 Input Power Requirement - The test set shall meet all applicable requirements of Specification MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amount.

(a) AC Power (Single Phase)

Operating Voltage Limits 115 ± 10% V ac, 400 ± 20 cps, 200 V A

(b) DC Power 28 Volt System

Operating Voltage Limits 27 ± 2 V dc

3.2.11.1 Voltage Protection - The equipment shall not be damaged when exposed to transient voltages as defined in Figure 2 of MIL-STD-704, and shall automatically resume normal operation when the voltage returns within limits of normal operation.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Description - The Test Harness, Direction Finder Set, AN/ALM-28(*) shall enable shop personnel to functionally test aircraft Direction Finder Group, AN/ALD-2 components for satisfactory operation before being installed in the aircraft. The test equipment shall consist of a Test Bench Harness which is capable of locating malfunction and making complete system check of the Direction Finder Group. By use of the test set, the operator shall be capable of locating malfunctions of any section of the Group to the module level. The test set accurately coordinates all inputs to the Group and functionally tests all associated switching as specified.

3.3.3 Operating Power Requirements - The test set shall accept 25 to 29 volts DC through receptacle J609 and 104 to 127 volts single phase, 380 to 420 cps AC through receptacle J605. This is the complete power requirements of the test set and aircraft components connected to it.

3.3.4 Grounding - The test set, under standard conditions shall not require any external grounding connections other than supplied with associated cables.

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3.3.5 Test Signal Leakage - The radiated leakage of the RF test signals shall not exceed 10 microvolts per meter (20 db above 1 microwatt) at 30 CM under the following conditions:

(a) RF output frequency adjusted to any frequency within the specified range.

(b) RF output adjusted to maximum.

(c) RF output connectors capped.

(d) Other equipment controls adjusted to any value within their operating range.

3.3.6 Oscillation - The test set, under standard operating conditions shall not cause any oscillation, fluctuation or variation not consistent with required test procedure.

3.4 Detail Requirements -

3.4.1 Case, Test Set CY-(*)/ALM-28 - The combination case shall be provided with a detachable cover in accordance with Specification MIL-T-21200 and shall contain items listed in 3.4.2 thru 3.4.6.

3.4.1.1 Function - The test set case shall be a portable carrying case providing housing facilities for the Test Set AN/ALM-28 and the necessary cable assemblies. (1 thru 27) See Paragraph 3.4.16.

3.4.1.2 Form Factor - The maximum dimensions of the test set case shall be 14 7/8 inches high, by 14 inches wide, by 18 3/4 inches long.

3.4.1.3 Weight - The weight of the test set case including cover and contents shall not exceed 50 pounds.

3.4.1.4 Cover CW-(*)/ALM-28 - The cover shall be in accordance with MIL-T-21200 and shall contain cables and antenna mounting panel.

3.4.2 Control-Indicator C-3622/ALM-28 - The Control - Indicator shall meet the following requirements:

3.4.2.1 Function - The Control-Indicator shall act as the central control panel, central indication panel and provide a means of checking aircraft components of the Direction Finder Set for functional operation.

3.4.2.2 Form Factor - The Control-Indicator circuitry is enclosed in the test set case and contains all the controls and indicators required for test harness operation. The maximum dimensions of the Control-Indicator as a unit shall not exceed 5 1/2 inches high, by 14 inches wide, by 19 inches long.

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3.4.2.3 Weight - The weight of the Control-Indicator shall not exceed 9 pounds.

3.4.2.4 Contents of Unit - The Control-Indicator shall contain the following circuits:

(a) AC and DC Power Circuits

(b) Antenna Selector Switch

(c) PORT-STBD Selector Switch

(d) Phone Jack

(e) Cable Receptacles

3.4.2.4.1 AC and DC Power - Primary power for the system under test is fed to the control panel through plugs P801 (28 volts, DC) and P802 (115 volts, 400 cps). Provide AC and DC power control and power distribution for the test harness and connected components.

3.4.2.4.2 Antenna Selector Switch - Permit a control panel selection of an antenna, or set of antennas paralleling the selection normally made by the operator from the Receiver Indicator.

3.4.2.4.3 PORT-STBD Selector Switch - Provide an indication that the proper antenna selection relays have been energized.

3.4.2.4.4 Phone Jack - Provide a jack for monitoring the audio output of Receiver Indicator.

3.4.2.5 Controls and Indicator - All controls and indicators are mounted on the front panel of the Control-Indicator.

Control or Indicator	Reference Designation	Function
28 V DC (Switch)	S1	Applies 28 volts DC to equipment when in ON position
28 V DC (Indicator)	DS1	Indicates, when lit, that 28 volts DC is applied to equipment.
115 V-400CPS (Switch)	S2	Applies 115 volts, 400cps, to equipment when in ON position.

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OPER (Indicator)	DS3	Indicates, when lit, that video signal level is below threshold setting value
INOPER (Indicator)	DS4	Indicates, when lit, that video signal level is above threshold setting value
PHONES (Jack)	J1	Connection Jack for monitoring Receiver Indicator Audio Output signal
STBD-PORT (Switch)	S3	Selects either Starboard or Port video in-out circuitry
Selector (Switch)	S4	Selects antenna, or combination of antennas for test
FWD (Indicators)	L DS6 S DS7 X DS8	Indicates, when lit, which forward antenna(s) has (have) been selected
AFT (Indicators)	L DS9 S DS10 X DS11	Indicates, when lit, which aft antenna(s) has (have) been selected

3.4.2.6 Electrical Connections - The test set shall be provided with the following connectors mounted on the front panel of the Control-Indicator and or the front panel of the Antenna Mounting Panel.

Reference Designation	Connector Type	Function
J601	MS3102A28-17P	To Connect cable W6
J602	MS3102A20-7PW	To Connect cable W2
J603	MS3102A18-8P	To connect cable W8

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J604	MS3102A24-27S	To Connect cable W7
J605	MS3102A14S-2P	To Connect cable W9
J606	MS3102A20-7P	To Connect cable W1
J607	MS3102A16S-1S	To Connect cable W3
J608	MS3102A24-28S	To Connect cable W5
J609	MS3102A14S-9P	To Connect cable W10
J701	UG-262B/U	To Connect cable W11
J702	UG-262B/U	To Connect cable W12
J703	UG-262B/U	To Connect cable W13
J704	UG-262B/U	To Connect cable W14
J705	UG-262B/U	To Connect cable W15
J706	UG-262B/U	To Connect cable W16
J707	67-02E12-7P or equal	To Connect cable W7
J708	UG-262B/U	To Connect cables W11 thru W16

3.4.3 Mounting Panel, Antenna MT-2474/ALM-28 - The antenna mounting panel, located in the cover of the combination case, is hinged at the bottom to a jack panel. Behind the mounting panel are mounting and storage facilities for the Direction Finder Set component mounts and interconnecting cable assemblies. Access to the area behind the mounting panel is accomplished by unlatching the panel and pulling it outboard and down to a horizontal position.

3.4.3.1 Meter - Shall provide a facility for determining the operating ability of the Starboard and Port Antenna groups by monitoring the bias current.

3.4.4 Mounting, Amplfier MT-2475/ALM-28 - The Amplifier mount provides mounting for the Amplifier during a bench test.

3.4.5 Mounting, Antenna Relay Box MT-2476/ALM-28 - The Antenna Relay Box Mount provides mounting for the Antenna Relay Box during a bench test.

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3.4.6 Mounting, Crystal Tester MT-2477/ALM-28 - The Crystal Tester Mount provides mounting for the Crystal Tester during a bench test.

3.4.7 Case, Test Adapter CY-3844/ALM-28 - The combination case shall be provided with a detachable cover in accordance with MIL-T-21200 and shall house the items listed in 3.4.8 through 3.4.15

3.4.7.1 Function - The Test Set Case CY-3844/ALM-28 shall be a portable carrying case providing housing facilities for the seven (7) Adapters, Test and the Cable Assembly, Special Purpose, Electrical, which are used to perform dynamic tests on the individual modules of Receiver-Indicator R-976/ALD-2 and R-976B/ALD-2.

3.4.7.2 Form Factor - The maximum dimensions of the case shall be 18 3/4 inches long, by 15 inches wide, by 11 inches high.

3.4.7.3 Weight - The weight of the test set case including cover shall not exceed 32 pounds.

3.4.7.4 Cover, CY-(*)/ALM-28 - The cover shall be in accordance with MIL-T-21200 and shall contain cables and soft wear.

3.4.8 Adapter, Test MX-4654/ALM-28 - The Adapter shall provide a means of connection to the High Voltage Power Supply PS-501 module of Indicators R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the indicator.

CAUTION: Extremely high voltage appears at test jacks. It is required that high voltage insulated test leads be used when performing voltage measurements.

3.4.8.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at Test Points A, D, H, C and B of High Voltage Power Supply PS-501 module of Indicators R-976/ALD-2 and R-976B/ALD-2. Five (5) test points on the Test Adapter shall permit dynamic testing of the module under test for fault isolation.

3.4.8.2 Form Factor - The over-all dimensions of the Adapter shall be 4 inches long by 3 inches wide by 4 inches high.

3.4.8.3 Weight - The weight of the Adapter shall be approximately three (3) pounds.

3.4.9 Adapter, Test MS-4655/ALM-28 - The Adapter shall provide a means of connection to the Video Amplifier AR-301 module of Indicator R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the indicator. (With W27 Cable).

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3.4.9.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at Test Points 1, 3, 5, 7, 9, 11, 13 and 19 of Video Amplifier AR-301 module of Indicator R-976/ALD-2 and R-976B/ALD-2. Eight (8) test points on the Test Adapter shall permit dynamic testing of the Video Amplfiier AR-301 module to determine fault isolation.

3.4.9.2 Form Factor - The over-all dimensions of the Adapter shall be 6 inches long, by 4 inches wide, by 1 inch high.

3.4.9.3 Weight - The weight of the Adapter shall be approximately three (3) pounds.

3.4.10 Adapter, Test MX-4656/ALM-28 - The Adapter shall provide a means of connection to the Low Voltage Power Supply PS-901 module of Indicator R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the Indicator.

3.4.10.1 Function - The Adapter, shall provide facilities for measuring voltage amplitudes at Test Points 1, 4, 11, 13 and 19 of Low Voltage Power Supply PS-901 module of Indicator R-976/ALD-2 and R-976B/ALD-2. Five (5) test points on the Test Adapter shall permit dynamic testing of the Low Voltage Power Supply module to determine fault isolation.

3.4.10.2 Form Factor - The over-all dimensions of the Adapter shall be 3 inches long, by 3 inches wide, by 1¼ inches high.

3.4.10.3 Weight - The weight of the Adapter shall be approximately two and one half (2½) pounds.

3.4.11 Adapter, Test MX-4657/ALM-28 - The Adapter shall provide a means of connection to the Pulse Amplifier AR-1001 module of Indicators R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the indicator.

3.4.11.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at Test Points 5 and 12 of Pulse Amplifier AR-1001 module of Indicators R-976/ALD-2 and R-976B/ALD-2. Two (2) test points on the Test Adapter shall permit dynamic testing of the Pulse Amplifier AR-1001 module to determine fault isolation.

3.4.11.2 Form Factor - The over-all dimensions of the Adapter shall be 2 3/4 inches long, by 2 inches wide, by 4¼ inches high.

3.4.11.3 Weight - The weight of the Adapter shall be approximately two and one half (2½) pounds.

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3.4.12 Adapter, Test MX-4667(*)/ALM-28 - The Adapter shall provide a means of connection to the Front Panel Assembly of the Receiver-Indicators, R-976/ALD-2 and R-976B/ALD-2, with the Front Panel Assembly, removed from the main chassis of the Receiver-Indicator.

3.4.12.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at test points 2 thru 8 of front Panel Assembles of Indicators R-976/ALD-2, and 2 thru 18 of R-976B/ALD-2.

3.4.12.2 Form Factor - The over-all dimensions of the Adapter shall be approximately 10 inches long, by 5 inches wide, by 2½ inches high.

3.4.12.3 Weight - The weight of the Adapter shall be approximately three (3) pounds.

3.4.13 Adapter, Test MX-4658/ALM-28 - The Adapter shall provide a means of connection to the Audio Amplfier AR-701 module of Indicators R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the indicator.

3.4.13.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at test points 3, 9, 10 and 11 of Audio Amplifier AR-701 module of Indicators R-976/ALD-2 and R-976B/ALD-2. Four (4) test points on the Test Adapter shall permit dynamic testing of the Audio Amplifier AR-701 module to determine fault isolation.

3.4.13.2 Form Factor - The over-all dimensions of the Adapter shall be 2 3/4 inches long, by 4 inches wide, by 4 inches high.

3.4.13.3 Weight - The weight of the Adapter shall be approximately three (3) pounds.

3.4.14 Adapter, Test MX-4659/ALM-28 - The Adapter shall provide a means of connection to the Aduio Amplfier AR-801 module of Indicators R-976/ALD-2 and R-976B/ALD-2, with the module removed from the main chassis of the indicator.

3.4.14.1 Function - The Adapter shall provide facilities for measuring voltage amplitudes at Test Points 3, 4, 6, 12 and 15 of Audio Amplfier AR-801 module of Indicators R-976/ALD-2 and R-976B/ALD-2. Five (5) test points on the Test Adapter shall permit dynamic testing of the Audio Amplifier AR-801 module to determine fault isolation.

3.4.14.2 Form Factor - The over-all dimensions of the Adapter shall be 2 3/4 inches long, by 4 inches wide, by 4 inches high.

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3.4.14.3 Weight - The weight of the Adapter shall be approximately three (3) pounds.

3.4.15 Cable Assembly, Special Purpose, Electrical CX-8648/ALM-28 - This Cable Assembly, Special Purpose, shall provide capabilities to couple power and signal sources from the main chassis of the indicator to the indicator mount, while performing maintenance work on the indicator modules.

3.4.15.1 Form Factor - Cable Assembly, Special Purpose shall be approximately 38 inches in over-all length and 36 inches long excluding terminations by 0.500 inch over-all cross section of cable.

3.4.15.2 Weight - The approximate weight of the Cable Assembly, Special Purpose shall not exceed one (1) pound.

3.4.16 Cables - The test set cables shall meet the following requirements: Housed in cover of case CY (*)ALM-28

3.4.16.1- Function - The test set cables shall provide a means for supplying input power to the test set and interconnections for the various associated equipments to the test set.

3.4.16.2 Form Factor - The dimensions of the test set cables shall be as follows:

Reference Designation	Function	Length (+3 Inches)
W1 thru W8	Special Purpose Electrical	72 inches
W9 and W10	Power, Electrical	72 inches
W11 thru W16	Video Frequency	48 inches
W17 thru W20	Video Frequency	72 inches
W21 thru W26	Radio Frequency	18 inches
W27 (CY-3844/ALM-28)	Video Signal Transfer	15 inches

3.4.16.3 Electrical Connections - Interconnecting cable plug numbers are as follows:

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Plug Numbers	Reference Designation	Plug Number
MS3106A-20-7P	W1	MS3106A-20-7S
MS3106A-20-7PW	W2	MS3106A-20-7W
MS3107A-18-24S	W3	MS3106A-16-1P
MS3107A-14-12S	W4	MS-3107A-14-12P
MS3106A-24-28P	W5	DM9702-27S
MS3106A-28-17S	W6	MS3106A-28-17P
MS-3106A-24-27P	W7	MS3107A-12-7S
MS3106A-18-8S	W8	MS3107A-18-24P
MS3106A-14-2S	W9	MS3106A-20-4P
MS3106A-14S-9S	W10	MS3106A-16-11P
UG-260-B/U	W11 thru W20	UG-260-B/U
UG-260-B/U	W21 thru W26	UG-262-B/U
MICRODOT INC.	W27	No. 93-3904 (93 ohm) or equal

3.5 Associated Equipment.- The Equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Antenna	AT-951/ALD-2
Antenna	AT-952/ALD-2
Antenna	AT-953/ALD-2 or AS-1355/ALD-2
Amplifier	AM-2509/ALD-2 or AM-2509B/ALD-2
Antenna Relay Box	RE-471/ALD-2 or RE471B/ALD-2
Crystal Tester	TS-1440/ALD-2
Receiver-Indicator	R-976/ALD-2 or R-976B/ALD-2
Mount, Receiver Ind.	MT-2284/ALD-2 or MT-2284A/ALD-2

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3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (See paragraph 6.7)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Proproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6 The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required test shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

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4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Proproduction test shall include environmental tests in accordance with the procedures of Specification MIL-T-21200 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Proproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

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4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the furnished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3, hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments slected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling test shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling.

4.4.3.1 Special Test Schedule - Selection of equipment for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test proceduces previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

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4.4.4 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure (s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test samples shall be selected by the government inspector in accordance with paragraph 6.12. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, or Special Tests may be selected for life tests.)

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. The record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life test shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity for the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, includin additional tests, when deemed necessary.

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4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the orginal shall be furnished the government inspector.

5.1 General - All major units and parts of the equipment shall be preserved, packages, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Test Equipment shall be housed in two combination cases and electrically connected to the equipment noted in paragraph 3.5. By means of interconnecting cables the test equipment shall be capable of checking the AN/AID-2 & 2B Systems.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 28 ± 1.0 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically, sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of Applicable Levels of packaging and packing (See 5.1)
(3) Data Requirements (See 3.6)

6.8 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.9 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Case, Test Set	CY-(*)/AIM-28	3.4.1.
Cover, Test Set	CY-(*)/AIM-28	3.4.1.4
Control Indicator	C-3622/AIM-28	3.4.2

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Continued

Mounting Panel, Antenna	MT-2474/ALM-28	3.4.3
Mounting Amplifier	MT-2475/ALM-28	3.4.4
Mounting Antenna Relay Box	MT-2476/ALM-28	3.4.5
Mounting Crystal Tester	MT-2477/ALM-28	3.4.6
Case, Test Adapter	CY-3844/ALM-28	3.4.7
Cover, Test Adapter	CW-(*)ALM-28	3.4.7.4
Adapter, Test	MX-4654/ALM-28	3.4.8
Adapter, Test	MX-4655/ALM-28	3.4.9
Adapter, Test	MX-4656/ALM-28	3.4.10
Adapter, Test	MX-4657/ALM-28	3.4.11
Adapter, Test	MX-4667/ALM-28	3.4.12
Adapter, Test	MX-4658/ALM-28	3.4.13
Adapter, Test	MX-4659/ALM-28	3.4.14
Cable Assembly Special Purpose, Electrical	CX-8648/ALM-28	3.4.15
Cable Assemblies, Harness	W-1 through W-27	3.4.16

6.9.1 Associated Equipment - The equipment shall operate with the following associated equipment which will not be supplied with the test equipment.

Antenna	AT-951/ALD-2
Antenna	AT-952/ALD-2
Antenna	AT-953/ALD-2 or AS-1355/ALD-2

6.10 Applicable Documents -

SPECIFICATIONS

Military

MIL-W-76	Wire and Cable, Hookup Electrical, Insulated
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-7078	Cable, Power electrically, copper shield aircraft 600 Volts
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-H-15362	Harnesses, Test Bench, Avionic Equipment, Gen. Spec. for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control System, Gen. Spec. for

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS21993	Cable Assemblies, Power Electrical

6.10.1 Availability of Documents -

(1) When requested specification, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.11 The parentheses (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.12 Life Test Schedule -

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specion in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obg information on the use of this specification which will insure that suitable products can be procured with a m amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

FICATION

-T-81409(WP) TEST HARNESS, DIRECTION FINDER SET AN/ALM-28(*)

ZATION *(Of submitter)* | CITY AND STATE

ACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

AL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

S ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

MENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

MARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

TTED BY *(Printed or typed name and activity)* | DATE

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